Ryan Breich

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PERSONAL PROFILE

I'm studying Robotics Engineering at UWE Bristol and am predicted to graduate with First Class Honours in 2027. Through university and personal projects, I've gained practical experience building robotic systems, programming microcontrollers and troubleshooting hardware and software. I enjoy working through technical problems and understanding how different parts of a system work together. I'm looking to develop these skills in an engineering role where I can contribute to real projects while continuing to learn.

EDUCATION

University of the West of England (UWE Bristol)
BEng Robotics Engineering | September 2024 – June 2027
Predicted classification: First Class Honours

Relevant modules: Mathematical Modelling for Electronics and Robotics, Programming for Engineers (C), Robot Control Systems (MATLAB), Microcontrollers Applications, Introduction to Machine Vision.

TECHNICAL SKILLS

- Programming and embedded systems: C, Python and MATLAB; experience with STM32 and Arduino microcontrollers, sensor integration and motor control.
- Electronics and hardware: Circuit schematics, breadboarding, soldering, introductory PCB layout and interpreting component datasheets.
- Testing and debugging: Oscilloscope and multimeter use, hardware fault-finding, sensor testing and analysing results to identify problems.
- Engineering tools: GitHub, Fusion 360, technical documentation, project planning and risk assessments.

ENGINEERING PROJECTS & EXPERIENCE

Final-Year Project — Brain-Inspired Robot Navigation | 2026–Present

- Developing a wheeled robot to evaluate a brain-inspired navigation method under realistic sensor and motor errors.
- Comparing physical testing against simulation to investigate how well the navigation method performs under real-world conditions.

AI-Assisted Robotic Arm — University Group Project | 2026–Present

- Working with a team to develop a robotic arm that uses AI to automate Petri dish swabbing for laboratory sample collection.
- Programming the arm's movements in Python and supporting the integration of its control with the wider system.
- Helping coordinate project activities, including scheduling and risk assessments.

STM32 Micromouse — University Group Project

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- Developed a microcontroller-based robot using C and STM32 peripherals, integrating sensors and motor control for maze navigation.
- Worked with teammates to troubleshoot hardware and software issues and improve the robot's behaviour.

Autonomous Rover — Personal Project

- Designed and 3D-printed a rover chassis with suspension inspired by NASA's Perseverance rover.
- Integrated ultrasonic sensors with an Arduino and developed C code for autonomous obstacle detection and avoidance.
- Resolved motor-driver wiring issues through systematic fault-finding and testing, improving understanding of hardware and motor-control behaviour.

UWE-FSAI Driverless Racing Team — Perception | 2026–Present

- Beginning to explore stereo-camera systems for depth perception using Python as part of an autonomous vehicle project.
- Building familiarity with the team's shared codebase and collaborative development practices through GitHub.

WORK EXPERIENCE

Euro Xchange — Intern | July 2026–Present

- Diagnose and resolve technical issues involving PCs, printers and storage drives, and assist with setting up staff accounts.
- Write scripts to automate routine tasks within the company's exchange software.
- Process currency transactions accurately while following established procedures and checks.

Oakberry — Team Member | October 2025–May 2026

- Prepared customer orders and maintained service standards in a busy environment.
- Handled customer transactions and helped keep daily operations running smoothly.

Zara Home — Stockroom Assistant | July 2025–September 2025

- Supported stockroom operations by organising deliveries, handling stock and replenishing products.
- Worked with colleagues to maintain an organised workspace and support store operations.

Royal Mail — Warehouse Operative | November 2024–May 2025

- Processed and sorted mail and parcels during night shifts while meeting operational requirements.
- Worked efficiently as part of a team in a fast-paced warehouse environment.

ADDITIONAL INFORMATION

Ryan Breich

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- Member of the UWE Robotics Society.
- Interested in robotics, embedded systems, electronics and autonomous technologies.
- Portfolio: rryanbbreich.github.io